KAIZEN MATHS

# Lesson starter

**Line and Curve Intersections**

Circle-Line Intersections

Name _____

**S1**

Find the intersection points:

Circle: $(x + 2)^2 + (y - 0)^2 = 9$

Line: $y = -3x$

**S2**

Find the intersection points:

Circle: $(x - 1)^2 + (y - 0)^2 = 16$

Line: $y = -0.577x - 4.041$

**S3**

Find the intersection points:

Circle: $(x - 0)^2 + (y + 2)^2 = 9$

Line: $y = 3x + 3$

**S4**

Find the intersection points:

Circle: $(x - 0)^2 + (y - 0)^2 = 16$

Line: $y = 3x - 1$



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ALG-5F81-L1 · S

Show your working

KAIZEN MATHS

# Exit ticket

**Line and Curve Intersections**

Circle-Line Intersections

Name _____

**EA1**

Find the intersection points:

Circle: $(x - 2)^2 + (y + 2)^2 = 9$

Line: $y = -0.894x - 4.236$

**EA2**

Find the intersection points:

Circle: $(x - 0)^2 + (y - 2)^2 = 25$

Line: $y = -3x$



Show your working

KAIZEN MATHS

# Exit ticket

**Line and Curve Intersections**

Circle-Line Intersections

Name _____

**EB1**

Find the intersection points:

Circle: $(x - 1)^2 + (y - 2)^2 = 16$

Line: $y = -3x - 3$



Show your working

KAIZEN MATHS

# Exit ticket

**Line and Curve Intersections**

Circle-Line Intersections

Name _____

**EA1**

Find the intersection points:

Circle: $(x - 2)^2 + (y + 2)^2 = 9$

Line: $y = -0.894x - 4.236$

**EA2**

Find the intersection points:

Circle: $(x - 0)^2 + (y - 2)^2 = 25$

Line: $y = -3x$

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Show your working

KAIZEN MATHS

# Exit ticket

**Line and Curve Intersections**

Circle-Line Intersections

Name _____

**EB1**

Find the intersection points:

Circle: $(x - 1)^2 + (y - 2)^2 = 16$

Line: $y = -3x - 3$

ALG-5F81-L1 · EB

Show your working

# Teacher answers & working

**Line and Curve Intersections · Circle-Line Intersections**

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**S1**

Find the intersection points:

Circle: $(x + 2)^2 + (y - 0)^2 = 9$

Line: $y = -3x$

**Answer:**

Two intersection points: (0.535, -1.605) and (-0.935, 2.805)

**Working:**

Given equations

Circle: $(x + 2)^2 + (y - 0)^2 = 9$

Line: $y = -3x$

Substitute the line into the conic

This gives the quadratic:

$10x^2 + 4x - 5 = 0$

Use the discriminant

$D = b^2 - 4ac = 216.000$

Solve the quadratic to find x-values: 0.535, -0.935

Substitute each x-value into $y = -3x$ to find y.

Point 1: x = 0.535, y = -1.605, so (0.535, -1.605)

Point 2: x = -0.935, y = 2.805, so (-0.935, 2.805)

# Teacher answers & working

**Line and Curve Intersections · Circle-Line Intersections**

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**S2**

Find the intersection points:

Circle: $(x - 1)^2 + (y - 0)^2 = 16$

Line: $y = -0.577x - 4.041$

**Answer:**

One intersection point: (-1.000, -3.464)

**Working:**

Given equations

Circle: $(x - 1)^2 + (y - 0)^2 = 16$

Line: $y = -0.577x - 4.041$

Substitute the line into the conic

This gives the quadratic:

$1.333x^2 + 2.667x + 1.333 = 0$

Use the discriminant

$D = b^2 - 4ac = 0.000$

Solve the quadratic to find x-values: -1.000

Substitute each x-value into $y = -0.577x - 4.041$ to find y.

Point 1: x = -1.000, y = -3.464, so (-1.000, -3.464)

# Teacher answers & working

**Line and Curve Intersections · Circle-Line Intersections**

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**S3**

Find the intersection points:

Circle: $(x - 0)^2 + (y + 2)^2 = 9$

Line: $y = 3x + 3$

**Answer:**

Two intersection points: (-0.694, 0.919) and (-2.306, -3.919)

**Working:**

Given equations

Circle: $(x - 0)^2 + (y + 2)^2 = 9$

Line: $y = 3x + 3$

Substitute the line into the conic

This gives the quadratic:

$10x^2 + 30x + 16 = 0$

Use the discriminant

$D = b^2 - 4ac = 260.000$

Solve the quadratic to find x-values: -0.694, -2.306

Substitute each x-value into $y = 3x + 3$ to find y.

Point 1: x = -0.694, y = 0.919, so (-0.694, 0.919)

Point 2: x = -2.306, y = -3.919, so (-2.306, -3.919)

# Teacher answers & working

**Line and Curve Intersections · Circle-Line Intersections**

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**S4**

Find the intersection points:

Circle: $(x - 0)^2 + (y - 0)^2 = 16$

Line: $y = 3x - 1$

**Answer:**

Two intersection points: (1.561, 3.683) and (-0.961, -3.883)

**Working:**

Given equations

Circle: $(x - 0)^2 + (y - 0)^2 = 16$

Line: $y = 3x - 1$

Substitute the line into the conic

This gives the quadratic:

$10x^2 - 6x - 15 = 0$

Use the discriminant

$D = b^2 - 4ac = 636.000$

Solve the quadratic to find x-values: 1.561, -0.961

Substitute each x-value into y = 3x - 1 to find y.

Point 1: x = 1.561, y = 3.683, so (1.561, 3.683)

Point 2: x = -0.961, y = -3.883, so (-0.961, -3.883)

# Teacher answers & working

**Line and Curve Intersections · Circle-Line Intersections**

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## EA1

Find the intersection points:

Circle: $(x - 2)^2 + (y + 2)^2 = 9$

Line: $y = -0.894x - 4.236$

**Answer:**

One intersection point: (0.000, -4.236)

**Working:**

Given equations

Circle: $(x - 2)^2 + (y + 2)^2 = 9$

Line: $y = -0.894x - 4.236$

Substitute the line into the conic

This gives the quadratic:

$1.8x^2 = 0$

Use the discriminant

$D = b^2 - 4ac = 0.000$

Solve the quadratic to find x-values: 0.000

Substitute each x-value into $y = -0.894x - 4.236$ to find y.

Point 1: x = 0.000, y = -4.236, so (0.000, -4.236)

# Teacher answers & working

**Line and Curve Intersections · Circle-Line Intersections**

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## EA2

Find the intersection points:

Circle: $(x - 0)^2 + (y - 2)^2 = 25$

Line: $y = -3x$

**Answer:**

Two intersection points: (0.968, -2.905) and (-2.168, 6.505)

**Working:**

Given equations

Circle: $(x - 0)^2 + (y - 2)^2 = 25$

Line: $y = -3x$

Substitute the line into the conic

This gives the quadratic:

$10x^2 + 12x - 21 = 0$

Use the discriminant

$D = b^2 - 4ac = 984.000$

Solve the quadratic to find x-values: 0.968, -2.168

Substitute each x-value into $y = -3x$ to find y.

Point 1: x = 0.968, y = -2.905, so (0.968, -2.905)

Point 2: x = -2.168, y = 6.505, so (-2.168, 6.505)

# Teacher answers & working

**Line and Curve Intersections · Circle-Line Intersections**

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**EB1**

Find the intersection points:

Circle: $(x - 1)^2 + (y - 2)^2 = 16$

Line: $y = -3x - 3$

**Answer:**

Two intersection points: (-0.420, -1.739) and (-2.380, 4.139)

**Working:**

Given equations

Circle: $(x - 1)^2 + (y - 2)^2 = 16$

Line: $y = -3x - 3$

Substitute the line into the conic

This gives the quadratic:

$10x^2 + 28x + 10 = 0$

Use the discriminant

$D = b^2 - 4ac = 384.000$

Solve the quadratic to find x-values: -0.420, -2.380

Substitute each x-value into y = -3x - 3 to find y.

Point 1: x = -0.420, y = -1.739, so (-0.420, -1.739)

Point 2: x = -2.380, y = 4.139, so (-2.380, 4.139)